

National University of Singapore

CS1101S — Programming Methodology

AY2023/2024 Semester 1

Midterm Assessment

Time allowed: 1 hour 30 minutes

ANSWER SHEET

STUDENT NUMBER								
A								
U		(0)	(0)	(0)	(0)	(0)	(0)	(A) (N)
A		(1)	(1)	(1)	(1)	(1)	(1)	(8) (R)
HT		(2)	(2)	(2)	(2)	(2)	(2)	(E) (U)
NT		(3)	(3)	(3)	(3)	(3)	(3)	(H) (W)
		(4)	(4)	(4)	(4)	(4)	(4)	(J) (X)
		(5)	(5)	(5)	(5)	(5)	(5)	(L) (Y)
		(6)	(6)	(6)	(6)	(6)	(6)	(M)
		(7)	(7)	(7)	(7)	(7)	(7)	
		(8)	(8)	(8)	(8)	(8)	(8)	
		(9)	(9)	(9)	(9)	(9)	(9)	

(1) ☐ **A** ☐ **B** ☐ **C** ☐ **D** ☐ **E** ☐ **F** ☐ **G** ☐ **H**

(2) ☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(3) ☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(4)

(5)



(6)

```
function is_entry(key, dict) {
    return accumulate(
        dict );
}
```

(7)

```
function remove_entry(key, dict) {
    return filter(
        dict );
}
```

(8)

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(9)

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(10)

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(11)

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(12)

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(13)

```
function circular_left_shift(xs) {
  if (is_null(xs)) {
    return xs;
  } else {
    
  }
}
```

(14)

```
function circular_right_shift(xs) {
  if (is_null(xs)) {
    return xs;
  } else {
    
  }
}
```

(15)

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(16)

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(17)

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(18)

```
function max_tree(tree) {
    return is_null(tree)
```

?

:

?

:

}

(19)

☐ A☐ B☐ C☐ D☐ E☐ F☐ G☐ H

(20)

```
function replace_elem(old_e, new_e, tree) {
    return accumulate_tree(
```

,

,

,

);

}

(21)

☐ A☐ B☐ C☐ D☐ E☐ F☐ G☐ H

(22)

☐ A☐ B☐ C☐ D☐ E☐ F☐ G☐ H

(23)

☐ A☐ B☐ C☐ D☐ E☐ F☐ G☐ H

(24)

☐ A☐ B☐ C☐ D☐ E☐ F☐ G☐ H

— END OF ANSWER SHEET —